Smart Waste Sorting System

Instance Segmentation

Team

Team Member 1	*Team Member 2*	*Team Member 3*
Karthik.K	Bharadwaj.G	Jagadish.P

Description of the Task

Overview

This project addresses the real-world challenge of automated waste sorting using Instance Segmentation, a high-level computer vision task that requires detecting individual waste objects, classifying them by material type, and producing precise pixel-level masks for each instance. This spatial information enables downstream automation such as robotic arms or conveyor belt sorting systems to physically separate waste by category (recyclable vs. non-recyclable).

Why Instance Segmentation?

- Challenging scope: combines object detection, classification, and pixel-wise segmentation in a single model.
- Real-world applicability: recycling systems require exact object boundaries, not just bounding boxes.
- Manual annotation required: each waste item must be individually labeled with a polygon mask.
- Novel dataset: the 6 waste categories chosen are NOT present in the COCO pretraining data.

Task Formulation

Given an input RGB image, the model outputs:

- A set of detected instances with class labels (e.g., plastic_bottle, cardboard_box).
- A bounding box [x, y, w, h] per instance.
- A pixel-level binary segmentation mask per instance.

Method, PyTorch Code & Paper

Name of the Method / Model

Selected Method
Model: Mask R-CNN
Backbone: ResNet-50-FPN
Framework: PyTorch (via Detectron2)
Pretrained: COCO 2017 (80 categories - none overlap with our waste classes)

PyTorch Code Link (GitHub)

GitHub Repository - Detectron2
URL: https://github.com/facebookresearch/detectron2

Paper Name & Reference

Paper
Title: Mask R-CNN
Link: https://arxiv.org/abs/1703.06870

How Mask R-CNN Works

Mask R-CNN extends Faster R-CNN with a parallel mask prediction branch:

Component	**Role**
Backbone (ResNet-50 + FPN)	Extracts multi-scale feature maps from the input image
Region Proposal Network (RPN)	Proposes candidate object bounding boxes
RoIAlign	Extracts fixed-size features per proposal with sub-pixel accuracy (fixes RoIPool misalignment)
Classification Head	Predicts class label + refined bounding box per proposal
Mask Head (FCN)	Predicts a 28x28 binary mask per class for each RoI
Loss	$L_{total} = L_{cls} + L_{box} + L_{mask}$ (binary cross-entropy per pixel for masks)

Detectron2 Dataset Format

Detectron2 consumes COCO-style JSON annotations. Each training sample consists of:

- Input: RGB image (JPEG/PNG).
- Target: COCO JSON with fields: image_id, category_id, bbox [x,y,w,h], segmentation (polygon list), area, iscrowd.

Dataset Description

Objects to Detect

The following 6 waste categories are selected.

ID	Category	Description
1	Plastic Bottles	Drink containers with diverse shapes and sizes.
2	Paper Waste	Common classroom waste like sheets, notes, and printouts.
3	Food Waste	Cafeteria leftovers with irregular shapes for segmentation.
4	Aluminum Cans	Beverage cans with reflective surfaces for real-world variation.
5	Disposable Cups	Coffee and juice cups frequently found around campus.
6	Tissue/Napkins	Soft, crumpled waste commonly available in cafeterias and restrooms.

Data Format

Input

- Type: RGB images (3-channel).
- File format: JPEG / PNG.

Target / Annotation Format

- Format: COCO-style JSON (polygon segmentation + bounding box per instance).
- Segmentation: polygon $[[x_1,y_1, x_2,y_2, ...]]$ drawn around each waste item.
- Bounding box: derived from polygon as $[x_{min}, y_{min}$, width, height].
- Category: integer ID mapped to class name.

Data Sources

Source	Description
Real-world photos	Captured at recycling centres, waste bins using smartphones
TACO Dataset	Trash Annotations in Context - open-source waste dataset (http://tacodataset.org)
TrashNet	Open-source waste classification dataset adapted for segmentation
Total	After quality filtering and deduplication

Dataset Split

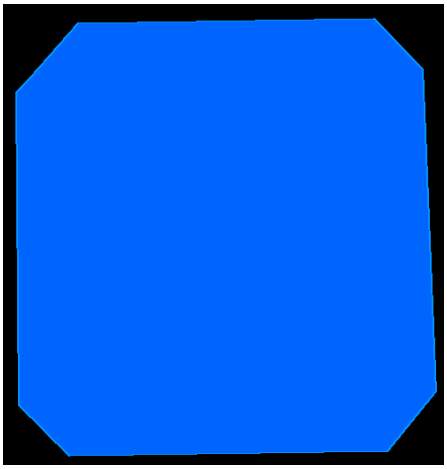
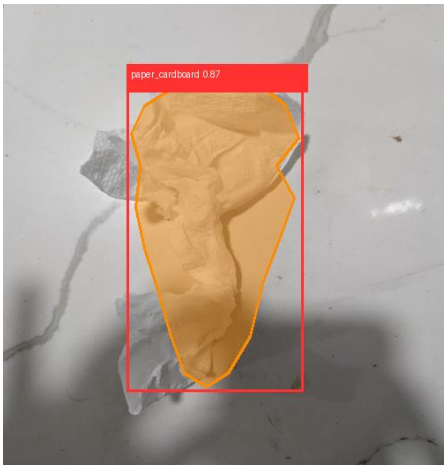
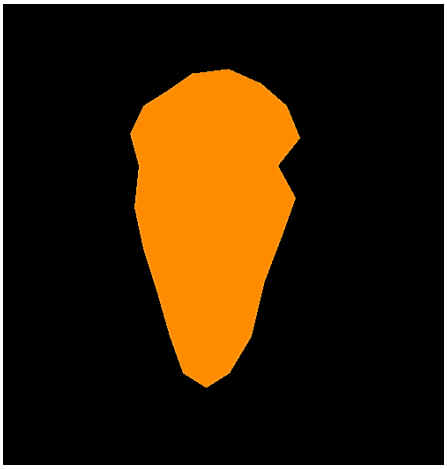
Split	Count	Purpose
Training	140 images (70%)	Model fine-tuning
Validation	30 images (15%)	Hyperparameter tuning, early stopping
Test	30 images (15%)	Final mAP evaluation

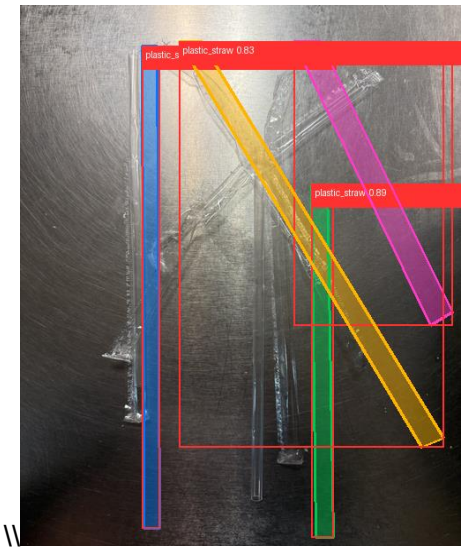
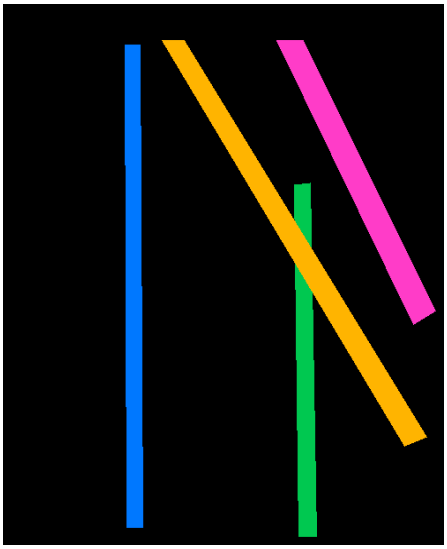
Annotation Tools

Primary Tool - CVAT (Computer Vision Annotation Tool)
URL: https://github.com/opencv/cvat
Type: Web-based, open-source
Features: Polygon annotation, instance masks, COCO JSON export, multi-annotator collaboration, quality review workflow.

- CVAT: Polygon drawing with direct COCO JSON export.
- LabelMe: Python-based quick annotation; output converted to COCO format.
- Roboflow (QA & Augmentation): Dataset quality review, augmentation pipeline, format conversion.

Sample Annotations

S.NO	Images	Segmentation Mask
1	**Image:1**	 **Mask:1**
2	 **Image:2**	 **Mask:2**

S.NO	Images	Segmentation Mask
3	 \\ **Image:3**	 **Mask:3**

Annotation Details

Sample 1: Styrofoam Food Container

Category	Styrofoam Box
Category ID	6
Bounding Box [x,y,w,h]	[15, 15, 475, 475]
Segmentation Polygon	[[30,80],[85,20],[420,15],[475,70],[490,420],[435,485],[75,490],[18,435],[15,95]]
Mask Color	Blue — RGB(0, 100, 255)
Instances in Image	1
Confidence Score	0.94
iscrowd	0
Location Captured	University Cafeteria

Sample 2: Paper Waste

Category	Paper cardboard
Category ID	3
Bounding Box	[140, 65, 200, 355]
Mask Color	Orange - RGB(255, 140, 0)
Instances in Image	1
Confidence Score	0.87
iscrowd	0
Location Captured	University Common Area
Segmentation Polygon	[[185,95], [215,75], [255,70], [290,85], [320,110], [335,145], [310,175], [330,210], [315,250], [295,300], [280,360], [255,400], [230,415], [205,400], [190,360], [175,310], [160,265], [150,220], [155,175], [145,140], [160,110]]

Sample 3: Plastic Straws

Category	Plastic Straw
Category ID	1
Total Instances	4 individual straws detected
Instance 1 BBox	[138, 45, 20, 535] - Mask: Blue RGB(0,120,255)
Instance 2 BBox	[330, 198, 25, 392]- Mask: Green RGB(0,200,80)
Instance 3 BBox	[310, 40, 180, 315] - Mask: Pink RGB(255,60,200)
Instance 4 BBox	[180, 40, 300, 450]- Mask: Yellow RGB(255,180,0)
Confidence Scores	0.92 / 0.89 / 0.85 / 0.83
iscrowd	0 (all instances)
Location Captured	University Cafeteria — metal tray surface

Links & References

Resource	Link
Paper - Mask R-CNN	https://arxiv.org/abs/1703.06870
PyTorch Code - Detectron2	https://github.com/facebookresearch/detectron2
Annotation Tool - CVAT	https://github.com/opencv/cvat
TACO Waste Dataset	http://tacodataset.org/
TrashNet Dataset	https://github.com/garythung/trashnet
COCO Format Specification	https://cocodataset.org/#format-data